Gravitational Dominance in Planetary Systems: An Energy-Based Hill-Radius Scaling Approach

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Abstract

This study examines gravitational dominance in planetary systems using an energy-based formulation. The relative influence of a celestial body and the Sun is quantified through a ratio of gravitational potential energies, evaluated as a function of distance normalized by the Hill radius to enable comparison across systems with widely different masses. The analysis is applied to Jupiter, Earth, Mars, and the asteroid 101955 Bennu. An energy-defined dominance region is found only in sufficiently massive systems: Jupiter exhibits a clear transition within a small fraction of its Hill radius, whereas Earth and Mars do not reach this threshold, and Bennu remains entirely solar-dominated. These results show that Hill-radius normalization does not yield a universal description of gravitational dominance and that energy-based measures provide a more restrictive and physically grounded interpretation.

Introduction

Gravitational influence in planetary systems is not determined by distance alone, but by the balance between competing gravitational fields. Identifying the region within which a celestial body exerts effective control over nearby particles is therefore a nontrivial problem, particularly in the presence of a much more massive central object.

A common reference for this problem is the Hill radius, which provides a characteristic length scale for gravitational influence (Murray & Dermott, 1999). It is widely used to estimate the region within which a body can retain satellites and maintain a degree of dynamical control. However, this definition is inherently geometric and does not directly describe the energetic conditions required for gravitational binding.

Comparing different systems introduces an additional complication. Planetary bodies span a wide range of masses, from giant planets to small asteroids, and this variation strongly affects gravitational interactions. Although distances can be normalized using the Hill radius, such scaling does not imply that gravitational dominance follows a common structure across systems. In particular, geometric scaling does not account for differences in the depth of the gravitational potential, limiting its ability to describe gravitational control in a physically consistent way.

This limitation motivates an alternative formulation based on energy. Rather than comparing local forces, gravitational dominance can be evaluated by comparing the potential energy contributions of the planetary body and the central star. In this study, dominance is quantified using an energy ratio evaluated as a function of normalized distance. The analysis is applied to Jupiter, Earth, Mars, and the asteroid 101955 Bennu, enabling comparison across systems with significantly different masses. The aim is to assess whether Hill-radius normalization yields a consistent description of gravitational dominance when interpreted through an energy-based framework.

2. Theoretical Background

2.1 Hill Radius

The Hill radius defines the characteristic scale at which the gravitational influence of a secondary body becomes comparable to that of a central mass in a three-body system. It is commonly used to approximate the region within which a body can retain satellites.

For a body of mass m orbiting a central mass M at a distance a, the Hill radius is

$$R_H = a\left(\frac{m}{3M}\right)^{1/3}.$$

This relation provides a geometric scaling for gravitational influence in planetary systems and serves as a reference for comparing systems with different masses and orbital distances (Murray & Dermott, 1999). In this study, distances are expressed in units of R_H.

2.2 Gravitational Potential Energy

Gravitational potential energy between two masses is given by

$$S_E = \frac{|U_{\mathrm{body}}|}{|U_{\mathrm{sun}}|}.$$

In a system consisting of a planetary body and a central star, a test particle experiences two contributions: the potential associated with the body, U_{body}, and that of the star, U_{sun}. Their relative magnitude determines which source dominates the binding.

Unlike force-based descriptions, which reflect local interactions, potential energy captures the global structure of the gravitational field and provides a direct measure of binding conditions.

2.3 Energy Ratio Definition

Gravitational dominance is quantified using the energy ratio

$$S_E = \frac{|U_{\text{body}}|}{|U_{\text{sun}}|}$$

This dimensionless quantity compares the contributions of the body and the star at a given location. The condition $S_E = 1$ defines the transition between body-dominated and star-dominated regimes.

By formulating the problem in terms of energy, the analysis focuses on the conditions required for gravitational binding rather than instantaneous force balance (Holman & Wiegert, 1999; Domingos et al., 2006).

3. Methodology

3.1 Systems Analyzed

This study examines four Solar System bodies spanning a wide range of masses: Earth, Mars, Jupiter, and the asteroid 101955 Bennu. These objects represent terrestrial planets, a gas giant, and a small-body system.

Using this set enables a controlled comparison across distinct regimes. Retaining the same systems as in the force-based analysis isolates the effect of reformulating the problem in terms of energy.

3.2 Distance Normalization

To enable meaningful comparison between systems with widely different physical scales, distances are expressed in normalized form using the Hill radius. The radial coordinate is defined as r/R_H, where R_H represents the characteristic scale of gravitational influence for each body.

This normalization expresses position relative to the extent of the body's gravitational region rather than in absolute units. Values of $r/R_H < 1$ correspond to locations within the Hill sphere, while larger values represent regions where the influence of the central star becomes increasingly significant. As a result, distances are interpreted in terms of relative gravitational context rather than physical separation.

Using this framework allows systems that differ by several orders of magnitude in mass and orbital size to be evaluated on a common basis. While normalization removes differences in spatial scale, it does not eliminate underlying physical differences between systems. Instead, it provides a consistent coordinate system in which variations in gravitational behavior can be compared directly.

3.3 Numerical Procedure

Gravitational influence is evaluated over normalized distances from $0.01 - 2\,R_H$, sampled using uniformly spaced points.

At each position, the gravitational potential associated with the body and the Sun is computed using

$$U = -\frac{GM}{r}$$

For the body, r is the radial distance. For the Sun, the distance is approximated by the orbital radius of the body, assuming that variations within the sampled region are small relative to this scale.

The energy ratio:

$$S_E = \frac{|U_{\mathrm{body}}|}{|U_{\mathrm{sun}}|}$$

is evaluated at each point.

3.4 Transition Criterion

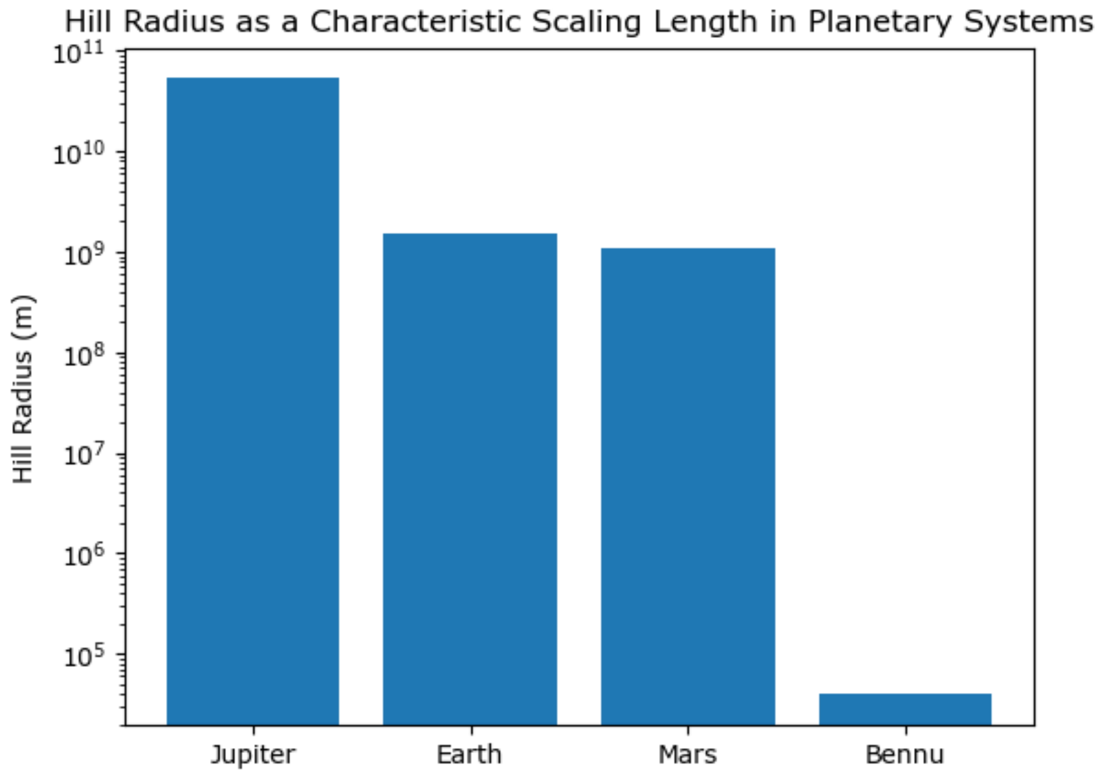
The transition between body-dominated and solar-dominated regimes is defined using the condition $S_E = 1$, corresponding to equal magnitudes of the gravitational potentials associated with the celestial body and the Sun. At this point, neither source provides a dominant contribution to the gravitational binding of the test particle.

In the numerical implementation, the transition point is identified as the first location at which the computed energy ratio crosses unity within the sampled domain. A small numerical tolerance is applied to account for finite sampling resolution and floating-point precision. This procedure ensures that the transition is determined consistently across all systems, allowing meaningful comparison of the presence or absence of energy-dominance regions.

4. Results

The results characterize gravitational dominance using the energy ratio S_E as a function of normalized distance r/R_H. This formulation enables direct comparison across systems with widely different masses and orbital scales. The following subsections present the computed profiles and identify the presence or absence of energy-dominance transitions for Earth, Mars, Jupiter, and the asteroid 101955 Bennu.

4.1 Hill Radius Comparison



Logarithmic comparison of the Hill radii R_H for Jupiter, Earth, Mars, and the asteroid 101955 Bennu, illustrating differences in gravitational scale across systems.

Figure 4.1 presents a logarithmic comparison of the Hill radii of Jupiter, Earth, Mars, and Bennu. The values span several orders of magnitude, from $\sim 10^{10}$ m for Jupiter to below $\sim 10^{5}$ m for Bennu.

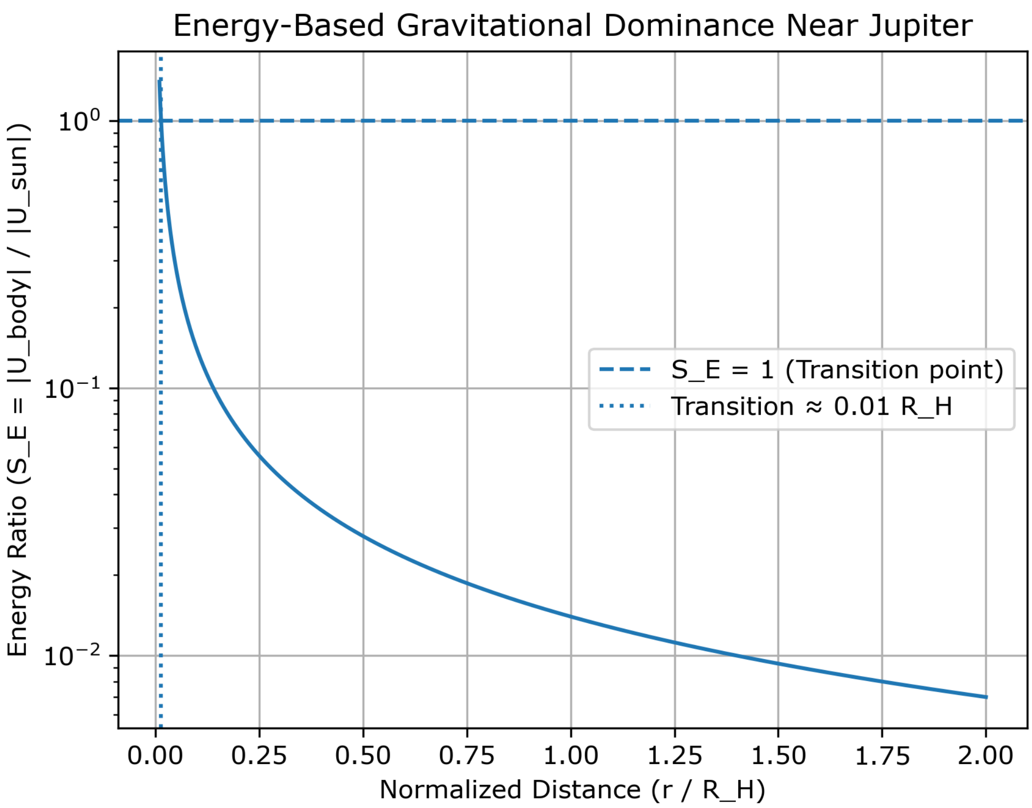
These differences reflect the distinct gravitational scales of the systems. While giant planets possess extended regions of influence, small bodies operate within a much more limited domain.

Because of this disparity, comparison in absolute distance is not meaningful. Expressing distance in units of the Hill radius provides a consistent basis for analyzing gravitational behavior across systems.

4.2 Energy Ratio Profiles

Figure 4.2 presents the energy ratio profiles for the analyzed systems as a function of normalized distance r/R_H. Although all curves decrease monotonically, they differ in whether the transition condition $S_E = 1$ is reached.

4.2.1 Jupiter

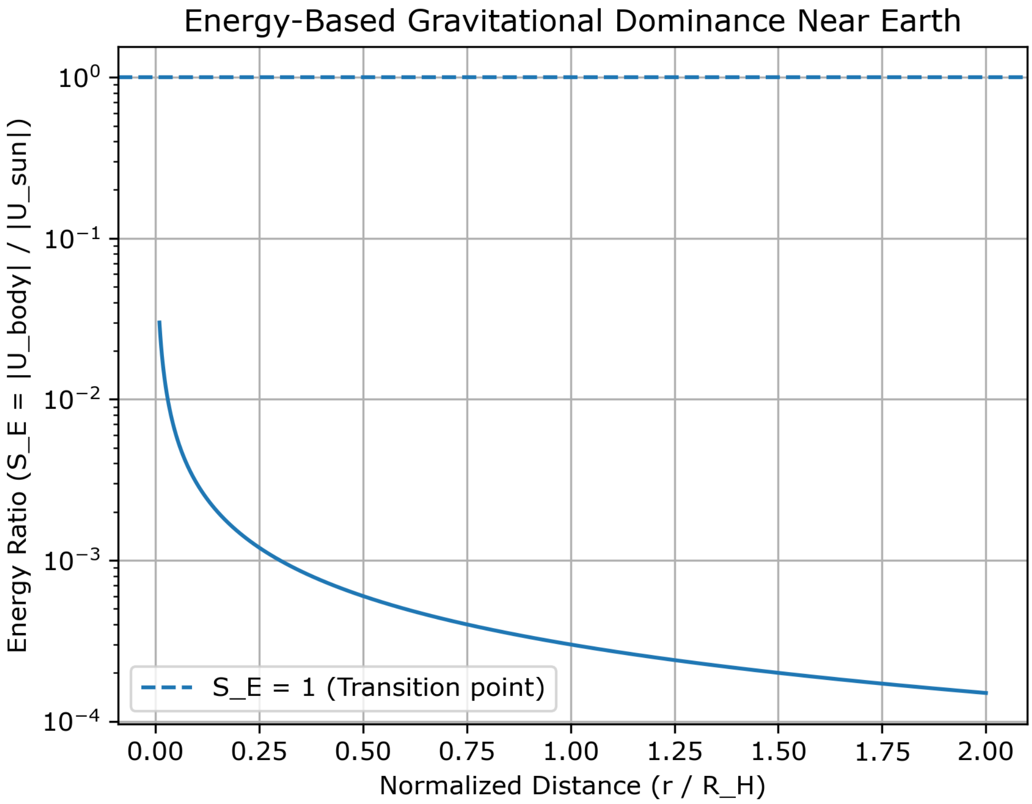


Energy ratio S_E as a function of normalized distance r/R_H for Jupiter. The intersection with $S_E = 1$ marks the transition between body-dominated and solar-dominated regimes.

Figure 4.2a shows $S_E\left(\frac{r}{R_H}\right)$ for Jupiter. A clear transition occurs at approximately $r \approx 0.01\,R_H$, where $S_E = 1$. For smaller radii, the energy ratio exceeds unity, while beyond this point it falls below unity.

The transition at a small fraction of the Hill radius highlights the difference between geometric and energetic measures of influence. Although the Hill radius defines a large dynamical region, the domain of energetic dominance is confined to its inner portion.

4.2.2 Earth

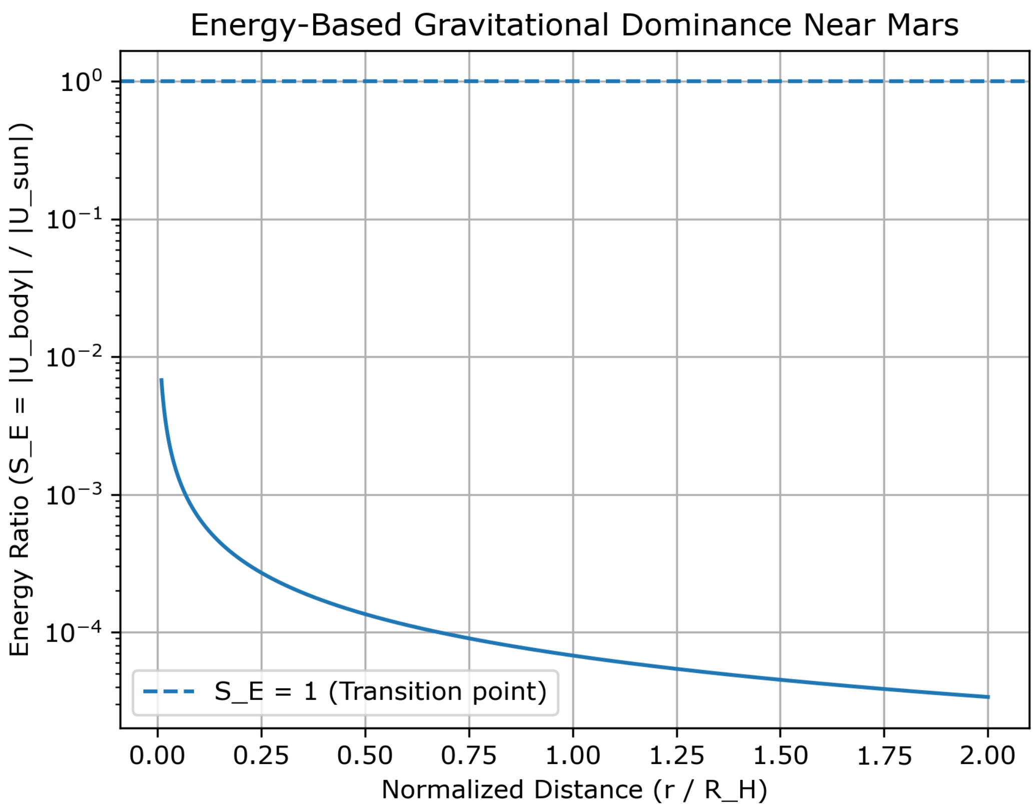


Energy ratio S_E as a function of normalized distance r/R_H for Earth.

Figure 4.2b shows $S_E\left(\frac{r}{R_H}\right)$ for Earth. The profile decreases monotonically and remains below unity across the entire domain.

No transition is observed, indicating that the Sun dominates energetically at all sampled distances.

4.2.3 Mars



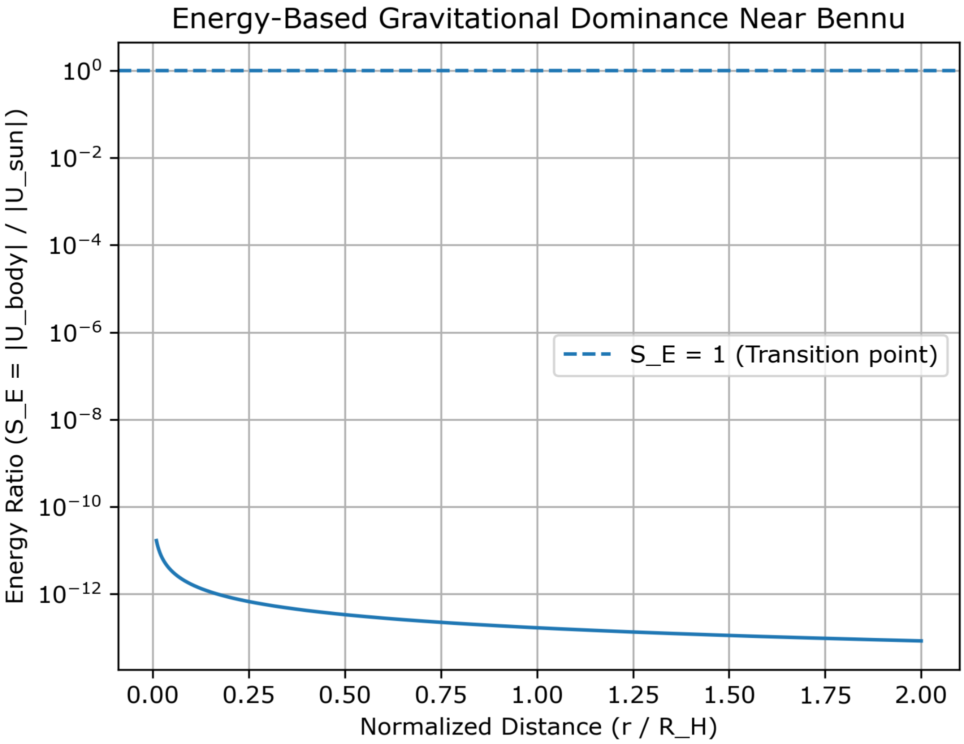
Energy ratio S_E as a function of normalized distance r/R_H for Mars.

Figure 4.2c shows $S_E\left(\frac{r}{R_H}\right)$ for Mars. The profile decreases monotonically and remains below unity across the entire domain, with no transition observed.

The behavior closely follows that of Earth, indicating the absence of an energy-dominant region within the sampled range.

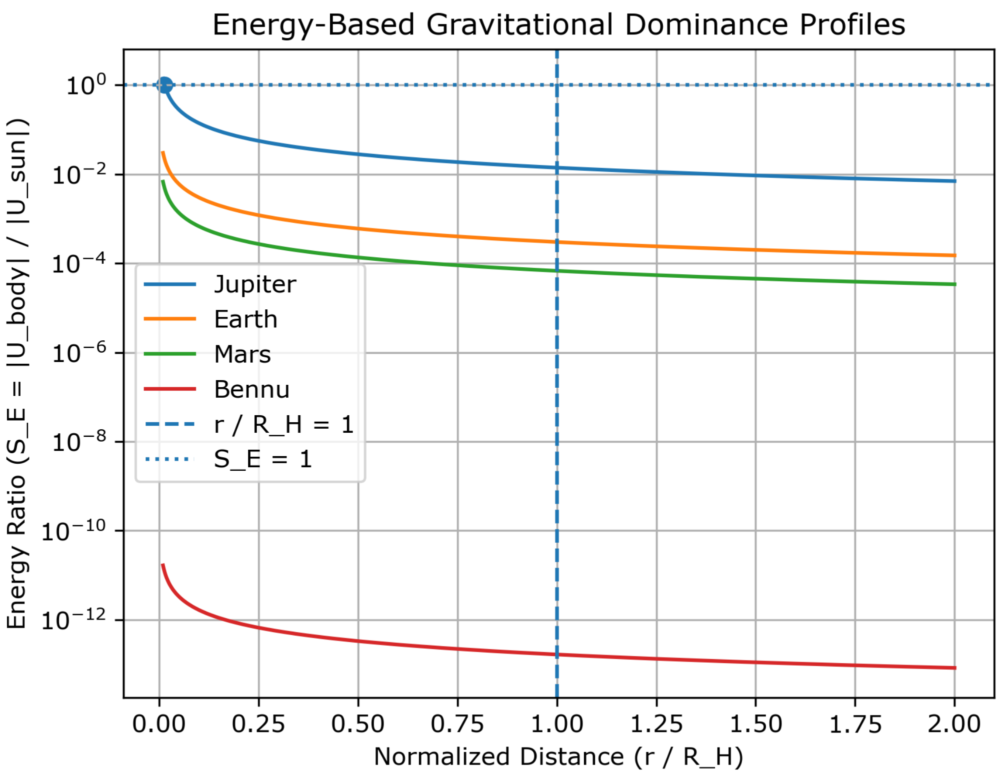
4.2.4 Bennu

Figure 4.2d shows $S_E\left(\frac{r}{R_H}\right)$ for Bennu. The profile follows the same decreasing trend but remains several orders of magnitude below unity throughout. No transition occurs at any distance, and the system remains solar-dominated across the full range.



Energy ratio S_E as a function of normalized distance r/R_H for Bennu.

4.3 Comparative Behaviour



Energy ratio S_E as a function of normalized distance r/R_H for Jupiter, Earth, Mars, and the asteroid 101955 Bennu. The horizontal line indicates the transition condition $S_E = 1$.

Figure 4.3 compares the profiles for all systems. Although the curves share a monotonic decrease, they do not collapse to a common structure under Hill-radius normalization.

Jupiter reaches the transition threshold, while Earth and Mars remain below it, and Bennu lies substantially lower. This ordering reflects a systematic dependence on mass: decreasing mass shifts the profile downward and removes the possibility of energetic dominance. Hill-radius scaling therefore provides a common spatial reference, but does not unify the energetic behavior across systems.

4.4 Transition Points

The transition point is defined by the condition $S_E = 1$, corresponding to equal contributions of gravitational potential from the celestial body and the Sun. This condition provides a reference for identifying the boundary between body-dominated and solar-dominated regimes in energetic terms.

Among the analyzed systems, only Jupiter exhibits a measurable transition within the sampled range, occurring at approximately $0.01 - 2\, R_H$. In contrast, no transition is observed for Earth or Mars, as their energy ratio profiles remain below unity across all sampled distances. Bennu similarly shows no transition, with values remaining several orders of magnitude below the threshold.

These results indicate that the existence of an energy-defined dominance boundary is not universal, but depends strongly on system mass. While the Hill radius provides a common geometric scale, the presence of an energetic transition emerges only in sufficiently massive systems.

5. Discussion

5.1 Physical Interpretation

Gravitational dominance, when evaluated using potential energy, differs fundamentally from its force-based interpretation. The Hill radius defines a geometric scale of influence, but does not correspond to a boundary of energetic control. This becomes evident when comparing the relative contributions of the planetary body and the Sun directly.

The results show that energetic dominance emerges only in sufficiently massive systems. Even for Jupiter, this region is confined to a small fraction of the Hill radius, while Earth, Mars, and Bennu remain energetically dominated by the Sun across the sampled range. This indicates that gravitational dominance is governed by potential depth rather than distance alone, and that the Hill radius should be interpreted as a scale of relevance rather than a region of guaranteed control.

5.2 Scaling Behavior

Hill-radius normalization provides a consistent spatial framework for comparing systems with different physical scales. Expressing distance as r/R_H allows the radial structure of gravitational influence to be examined across bodies that differ by orders of magnitude in mass and orbital size.

However, this normalization does not produce a universal energetic profile. While all systems exhibit a similar radial trend, their magnitudes remain distinct and do not collapse onto a common curve. This reflects the explicit dependence of the energy ratio on system mass, indicating that normalization removes geometric differences but does not eliminate the underlying physical scaling.

5.3 Broader Implications

The results imply that the Hill radius should not be interpreted as a universal boundary of gravitational dominance, but rather as a geometric reference whose meaning depends on system mass. In particular, the absence of energetically dominant regions for Earth and Mars shows that a Hill sphere does not necessarily correspond to a region of effective gravitational binding.

This distinction has broader implications for how gravitational environments are interpreted. Approaches based solely on spatial boundaries may overestimate the extent of planetary influence, especially in lower-mass systems. An energy-based framework provides a more restrictive and physically meaningful criterion for identifying regions of gravitational control.

5.4 Limitations

The analysis is based on a simplified energy ratio and does not include full orbital dynamics. It does not solve the restricted three-body problem or account for time-dependent effects such as velocity and perturbations.

As a result, the condition $S_E = 1$ should be interpreted as an approximate indicator rather than a precise stability boundary. A more complete treatment would require numerical N-body simulations capable of capturing long-term dynamical evolution and providing a direct connection between energetic dominance and orbital stability.

6. Conclusion

This study examined gravitational dominance in planetary systems using an energy-based formulation. By comparing the gravitational potentials of a celestial body and the Sun within a Hill-radius framework, the analysis provides a consistent basis for evaluating how dominance varies across systems of different mass.

The results show that an energy-defined dominance region is not a general feature of planetary systems, but emerges only in sufficiently massive cases. Hill-radius normalization captures spatial scaling, but does not yield a universal description of energetic behavior. In particular, the existence of a body-dominated region depends on the depth of the gravitational potential rather than on geometric considerations alone.

These findings indicate that the Hill radius should be interpreted as a reference scale rather than a boundary of gravitational control. While it remains useful for comparing systems, it does not guarantee the presence of an energetically bound region.

More broadly, the analysis highlights the importance of energy-based approaches in describing gravitational systems. Although simplified, the model provides a framework for understanding how gravitational dominance varies across different regimes and suggests that energetic criteria offer a more restrictive and physically meaningful description of gravitational influence.

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